

Introduction to

Databases & SQL



Databases



SQL Basics



Aggregations



Filtering & Sorting

No prior coding needed ✨ Simple words ✨ Lots of examples



What is a Database?



Think of it like...



Library

Books = Data, Shelves = Tables



Spreadsheet

But way more powerful!



Filing Cabinet

Organised folders of info



Why Use a Database?



Store thousands of records safely



Find any data in seconds



Avoid duplicate info



Multiple people can use it together



Great for data analysis!



FUN FACT: YouTube, Instagram, and Spotify all use huge databases!



Relational Databases: Tables, Rows & Columns



Table

Like a spreadsheet tab. Holds one type of data (e.g. Students)



Row

One single record — e.g. one student's full info



Column

One type of info — e.g. Name, Age, Grade



Example: Students Table

StudentID	Name	Age	Grade
1	Aanya Singh	10	A
2	Rohan Mehta	11	B+
3	Priya Joshi	10	A+

Each row = 1 student
Each column = 1 type of info
All stored in ONE table!



Relational = Tables can be LINKED together using a common column (like StudentID)!



Introduction to SQL

SQL = Structured Query Language

It's how we TALK to databases — we ask questions, and the database answers!

SELECT

Get / read data from a table

DELETE

Remove data from a table

INSERT

Add new rows of data

CREATE

Make brand new tables

UPDATE

Change existing data



SQL is almost like English — easy to read, write and understand!



Popular Database Systems (RDBMS)

RDBMS = Relational Database Management System — the software that runs your database!



MySQL

Best for:

Web apps & blogs

Popular for beginners



Powers WordPress & Facebook!



Best for Class!



PostgreSQL

Best for:

Complex projects

Advanced features



Free & very powerful!



For Big Projects



SQLite

Best for:

Mobile & small apps

Super lightweight



No server setup needed!



Quick Start



We'll use **MySQL + MySQL Workbench** — it has a nice visual interface, perfect for beginners!



Setting Up MySQL & MySQL Workbench

1

Download MySQL

Go to mysql.com/downloads
Choose MySQL Community Server — it's FREE!

2

Install MySQL

Run the installer & choose
'Developer Default'. It sets everything up!

3

Get MySQL Workbench

Workbench is a visual tool — like a
control panel for your database!

4

Connect & Start!

Open Workbench → click connection
→ New SQL file → write your first query!



MySQL Workbench lets you write SQL and see your results instantly!



SELECT, FROM & WHERE

💡 Think of SQL like asking: "Show me the NAMES of all STUDENTS who are in GRADE 10"

SELECT

What do you want to see?

SELECT name

FROM

Which table to look in?

FROM students

WHERE

Which rows to filter?

WHERE grade = 10



Full SQL Example:

```
SELECT name, age FROM students WHERE grade = 10;
```



This gets the name & age of every Grade 10 student!



Operators, ORDER BY & NULL Values

1 2
3 4

Comparison Operators

Symbol	Meaning	Example
=	Equals	age = 10
>	Greater than	age > 10
<	Less than	score < 50
>=	Greater or equal	grade >= 8
!=	Not equal	city != 'Delhi'



Logical: AND / OR / NOT

```
WHERE age > 10 AND grade = 'A'
```

abc

ORDER BY — Sort Results!

```
SELECT name, score  
FROM students  
ORDER BY score DESC;
```

ASC = smallest → biggest | DESC = biggest → smallest



What is NULL?

NULL = missing / unknown value. Not zero, not empty — just UNKNOWN!

```
WHERE phone IS NULL -- finds missing  
values
```




Aggregation Functions

Special functions that do MATH on your data automatically!



SUM()

Add all values together

```
SELECT SUM(score)
```



AVG()

Find the average value

```
SELECT AVG(age)
```



COUNT()

Count the total rows

```
SELECT COUNT(*)
```



MIN()

Find the smallest value

```
SELECT MIN(score)
```



MAX()

Find the biggest value

```
SELECT MAX(score)
```



GROUP BY — Summarise by Category

GROUP BY groups rows with the same value — then you can calculate totals, averages etc. per group!

Without GROUP BY — all rows

Name	Subject
Aanya	Maths
Rohan	Science
Aanya	Science
Rohan	Maths
Priya	Maths

✓ With GROUP BY — summarised

```
SELECT subject,
```

```
COUNT(*)
```

```
FROM scores
```

```
GROUP BY subject;
```

Result — one row per subject:

Subject	COUNT(*)
Maths	3
Science	2



GROUP BY collapses many rows into summary rows — great for reports & data analysis!



HAVING — Filter Your Groups

HAVING is like WHERE — but it filters GROUPS, not individual rows!

WHERE — filters rows first

```
SELECT grade, COUNT(*)  
  
FROM students  
  
WHERE age > 10  
  
GROUP BY grade;
```

WHERE runs BEFORE grouping.
Only age > 10 rows enter the group.

HAVING — filters groups after

```
SELECT grade, COUNT(*)  
  
FROM students  
  
GROUP BY grade  
  
HAVING COUNT(*) >= 30;
```

HAVING runs AFTER grouping.
Only groups with 30+ students show up.



WHERE → GROUP BY → HAVING (always in this order!)



What You've Learnt!



Databases

Organised stores of data



Tables, Rows & Columns

Structure of a database



SQL Language

How to talk to databases



MySQL, PostgreSQL, SQLite

Popular RDBMS tools



Setting Up MySQL

Install & use Workbench



SELECT / FROM / WHERE

Basic querying



Operators & ORDER BY

Filter & sort results



SUM, AVG, COUNT, MIN, MAX

Aggregation functions



GROUP BY & HAVING

Summarise grouped data



You are now ready to write your own SQL queries! Keep practising!

